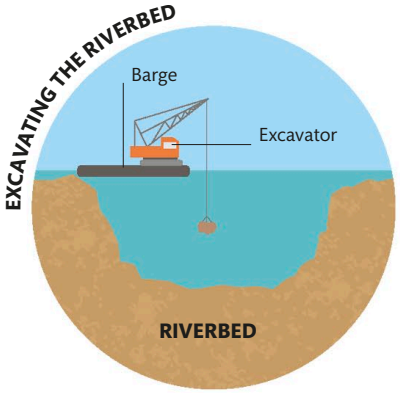
Tunnels

A tunnel is basically a large tube through earth or rock that has been reinforced to prevent collapse. Building tunnels usually requires specialized machinery.

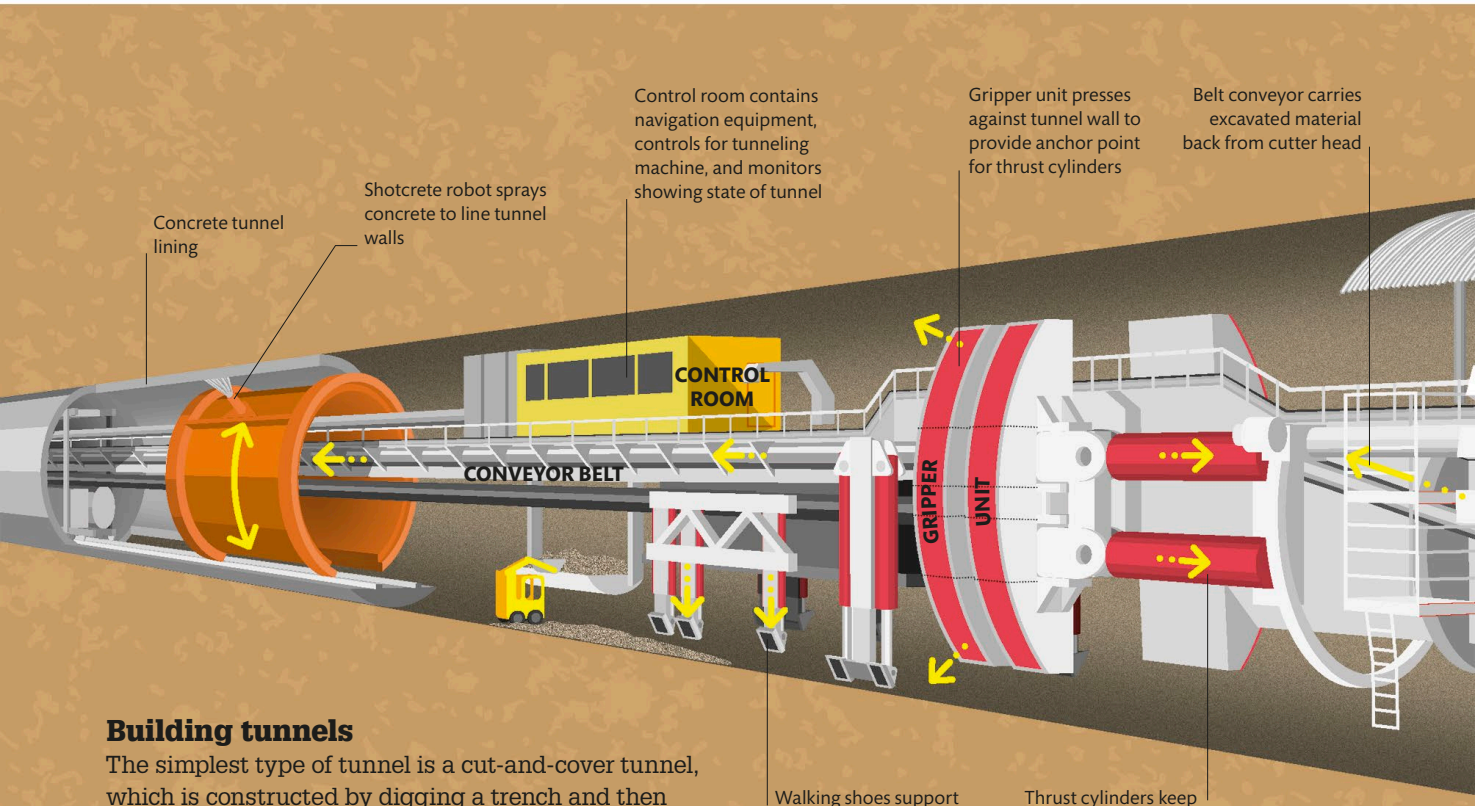
Underwater tunnels

Tunnels can be bored under bodies of water using a tunnel boring machine (see below). One example of a bored underwater tunnel is the Channel Tunnel, which connects Britain and France. However, it is often faster and more cost-effective to build tunnels under water using the immersed tube method.

Immersed tube tunnel
The immersed tube method involves making a tunnel in sections on land and then bringing the sections to the construction site, where they are submerged and joined to each other.



1 To reduce the risk of the tunnel interfering with shipping, a trench for the tunnel is dug into the bed of the river, lake, or sea using an excavator mounted on a barge.



Building tunnels

The simplest type of tunnel is a cut-and-cover tunnel, which is constructed by digging a trench and then covering it. A bored tunnel is dug through soil or rock, typically by using a tunnel boring machine (TBM), or "mole." When building long tunnels, it is often necessary to bore additional shafts to provide emergency escape routes and remove toxic fumes.